Reimagining Climate Model Development in the Age of AI

(and future perspectives for AWI)

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Who am I? Role of AI in my career path

- Postdoc / scientist in **climate dynamics section**
- Working on ocean model development: **FESOM**, km-scale coupled simulations.
- Degrees in **Atmospheric Science** and **Mechanical Engineering**
- No formal education in computer science / IT
- Just compulsive-broad interests: physics of climate, numerical algorithms, IT, hardware sensors. **Jack of many trades.**
- … many bad habits not suitable for a good scientific career.

… and then the AI era happened

- A potential cure for my bad habits and compulsions
- **Early adopter**, especially for model development
- Treated it as a student intern, recognised its value and was **happy to self-pay**
- Often **overwhelmed** with possibilities

But …

My political beliefs on AI

- Concerned about misuse against humanity.
- Trained on public data, but now called *their* model.
- Most data-privacy promises amount to pinky promises without an audit.

> But I believe that to change the rules, you need **skin in the game**.
>
> **Pick your battles** wisely, but don't lose sight of the broader goals.

AI in ocean model development: FESOM

Now

- ✓ Overworked developers: Features and Issues pending for years
- ✓ Very few tests and automation
- ✓ Inflexible architecture
- ✓ Performance portability: Modern CPUs, GPUs, Mixed precision
- ✓ No GPU port
- ✓ Reproducibility Issues

CORE2 mesh

160 → 400

SYPD

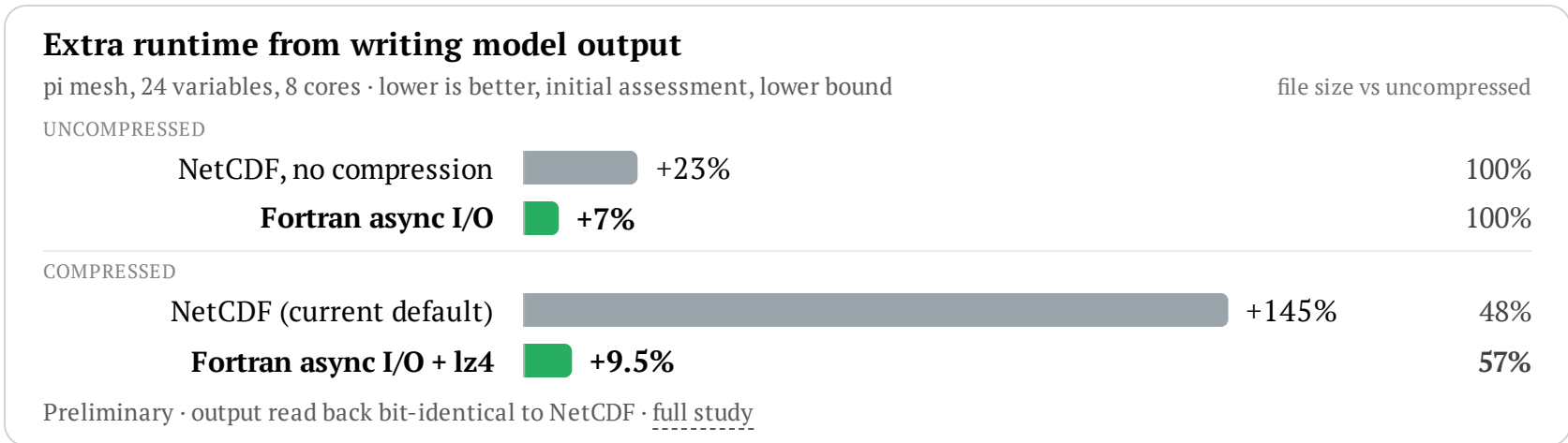
simulated years per day and SP

Goal: multiple millennia at coarse resolution, a few decades at km-scale resolution

> With a **small team of developers**, AI lets us **pioneer climate model development**.

Going wild with AI: for a greater good

- **Improve the (open-source) GNU compilers: A key piece of any software infrastructure**
  - Improve quality and performance on par with commercial compilers.
  - Climate model example: Asynchronous I/O in gfortran, plugged into FESOM's output
    - Part of the **standard Fortran interface**: every Fortran-based model benefits, no extra libraries



- **Differentiable ocean model**: a FESOM shadow in Python (@ Nikolay Koldunov)
- **Next adventure: build hardware** for efficient climate models, or even sensors for observations

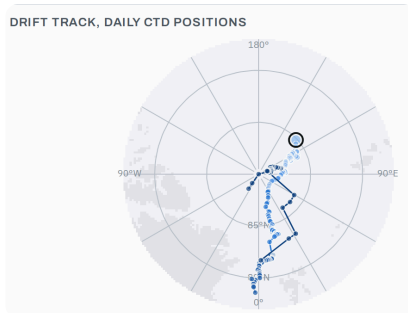
AI-enabled opportunities at AWI

Overcome past limitations: can we **constrain models with observations**, and **drive new targeted campaigns with models**?

- Regular **(sub) km-scale simulations** are within reach
- Model resolutions getting closer to observations.
- Tools to compare models and observations are faster to build.
- With a **differentiable model shadow**, can we constrain model **parametrisations**?

> Models and observations can finally **inform each other**: models constrained by data, campaigns guided by models.

Virtual MOSAiC Field Campaign



● Observed — IFS-FESOM2-SR — ICON-ESM-ER

EERIE runs sampled along the Polarstern drift · open interactive

AI-enabled opportunities at AWI : Summary

AWI is exceptionally well placed to advance Earth system modelling, unparalleled, especially **polar climates**

Observations

- Ocean and polar observations
- Bathymetry

Process expertise

- Sea ice
- Atmospheric boundary layers
- Marine biogeochemistry
- Ice sheets and glaciers
- Palaeoclimate

→ **Coupled model development**

> Leverage this expertise to build the **best climate model**: with AI, we can **overcome the past hurdles**.

Let's become **pioneers** in using **sustainable AI** as a **force for good in science** and not resist it.

Happy to take Comments, challenges and in-house consultations: suvarchal.cheedela@awi.de